

How to use this interactive PDF: Click **Show / Hide Answer** under any question to reveal or hide its answer. This requires **Adobe Acrobat Reader** (free). In other viewers the buttons may not work — use your viewer's **Layers** panel, or the **Reveal All** button below.

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CCNA Subnetting Drill Sheet

50+ Problems with Full Worked Answers — From Baby Steps to Exam Speed

Why this exists: Subnetting shows up ALL OVER the CCNA exam, and you have to do it **fast** (about 60–90 seconds per question). Reading about it isn't enough — you have to drill it until it's automatic, like times tables. This sheet gives you 50+ problems, grouped from easy to hard, each with a **step-by-step answer**.

How to use it: Cover the answer with your hand or a piece of paper. Work the problem on scratch paper FIRST. Then check. Do **5–10 problems every single day**. In two weeks you'll be fast.

Where this fits. The theory behind every method here lives in **Chapters 11–13 of the CCNA Study Guide** (IP addressing, subnetting, IPv6) — if a step doesn't make sense, that's where to look. Once these are automatic, the **Practice Question Bank** puts subnetting back into full exam-style questions.

The Toolkit (Everything You Need on One Page)

The Two Magic Charts

Chart 1 — Bit values in one octet (memorize left to right):

128	64	32	16	8	4	2	1
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Chart 2 — The subnet "cheat table" (memorize this cold):

CIDR	Mask (last octet)	Block Size	Hosts (usable)
/24	0	256	254
/25	128	128	126
/26	192	64	62
/27	224	32	30
/28	240	16	14
/29	248	8	6
/30	252	4	2

Notice the pattern: **Block size doubles as you go up** (4, 8, 16, 32, 64, 128, 256), and **Mask value** = 256 – block size.

The 5 Facts You Compute for Any Subnet

For any IP + mask, you can always find these 5 things:

1. **Block size** = 256 – (the interesting octet of the mask).
2. **Network address** = the block start the IP falls into (all host bits 0).
3. **Broadcast address** = one below the next network (all host bits 1).
4. **First usable host** = network + 1.
5. **Last usable host** = broadcast – 1.

The Formulas

```
Number of hosts per subnet = 2^(host bits) - 2
Number of subnets         = 2^(borrowed bits)
Host bits                  = 32 - CIDR
```

The 3-Step "Where Does This IP Live?" Method

1. Find the **block size** (256 – interesting mask octet).
2. Count up in the interesting octet by the block size (0, block, 2×block...) until you pass the IP's value. The one **just below or equal** is the **network**.
3. **Broadcast** = next network – 1. Hosts sit between them.

SECTION 1 — Warm-Up: Masks, CIDR & Host Counts

(Goal: instant recall. Aim for under 15 seconds each.)

Q1. Convert **/26** to a dotted-decimal mask.

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Q2. Convert **255.255.255.240** to CIDR.

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Q3. Convert /19 to a dotted-decimal mask.

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Q4. How many **usable hosts** are in a /27?

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Q5. How many **usable hosts** are in a /30?

Show / Hide Answer

Q6. How many **usable hosts** are in a /22?

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Q7. Convert binary 11000000 to decimal.

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Q8. Convert decimal 224 to binary.

Show / Hide Answer

Q9. What is the **block size** of a /29?

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Q10. A mask ends in .248. What CIDR is that, and how many hosts?

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SECTION 2 — Find the Network & Broadcast Address

(The bread-and-butter skill. Aim for under 60 seconds each.)

Q11. Given 192.168.1.77 /26, find the network and broadcast address.

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Q12. Given 10.0.0.200 /27, find the network and broadcast.

Show / Hide Answer

Q13. Given 172.16.5.130 /25, find the network and broadcast.

Show / Hide Answer

Q14. Given 192.168.10.44 /28, find the network and broadcast.

Show / Hide Answer

Q15. Given 10.10.10.10 /29, find the network and broadcast.

Show / Hide Answer

Q16. Given 192.168.1.100 /30, find the network and broadcast.

Show / Hide Answer

Q17. Given 172.16.99.200 /21, find the network and broadcast. (Now the interesting octet is the THIRD one!)

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Q18. Given 10.50.150.7 /20, find the network and broadcast.

Show / Hide Answer

Q19. Given 192.168.200.55 /18, find the network and broadcast.

Show / Hide Answer

Q20. Given 10.1.1.1 /8, find the network and broadcast.

Show / Hide Answer

SECTION 3 — Find the Usable Host Range

(Network + 1 to Broadcast – 1. Put it all together.)

Q21. For 192.168.1.77 /26, list the **first and last usable host**.

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Q22. For 172.16.5.130 /25, list the first and last usable host.

Show / Hide Answer

Q23. For 10.10.10.10 /29, list the first and last usable host.

Show / Hide Answer

Q24. For 192.168.1.100 /30, list the first and last usable host.

Show / Hide Answer

Q25. For 172.16.99.200 /21, list the first and last usable host.

Show / Hide Answer

Q26. Is 192.168.1.64 a **usable host** address in the 192.168.1.0 /26 scheme? Why?

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Q27. Is 192.168.1.191 usable in a /26 scheme? Why?

Show / Hide Answer

Q28. Two PCs: 192.168.1.70 /26 and 192.168.1.130 /26. Can they talk **directly** (same subnet)?

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Q29. Two PCs: 10.0.0.33 /27 and 10.0.0.60 /27. Same subnet?

Show / Hide Answer

Q30. A device is configured **192.168.5.31 /27**. What's wrong?

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SECTION 4 — "How Many Subnets / Hosts Do I Need?" (Design)

(These are the word problems. Read carefully!)

Q31. You have **192.168.1.0 /24** and need **4 equal subnets**. What mask and how many hosts each?

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Q32. You have **192.168.1.0 /24** and need **at least 50 hosts per subnet**, as many subnets as possible. What mask?

Show / Hide Answer

Q33. You need **at least 500 hosts** in one subnet. What is the smallest mask (largest CIDR number) that works?

Show / Hide Answer

Q34. You need **exactly 6 subnets** from **172.16.0.0 /16**, each as large as possible. What mask?

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Q35. How many **/30** subnets can you make from a single **/24**?

Show / Hide Answer

Q36. How many **/26** subnets fit in a **/22**?

Show / Hide Answer

Q37. A company needs 200 hosts in one LAN. Will a /25 work? What about a single /24?

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Q38. You need a subnet for a **point-to-point link between two routers** with no wasted addresses. What mask?

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Q39. From 10.0.0.0 /24, you subnet into /28s. What is the **network address of the 5th subnet**?

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Q40. From 192.168.1.0 /24 subnetted into /27s, what is the **broadcast of the 3rd subnet**?

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SECTION 5 — VLSM (Variable Length Subnet Masks)

(Different-sized subnets from one block. The GOLDEN RULE: always assign the biggest subnet first!)

Q41. From 192.168.1.0 /24, create subnets for: Sales (100 hosts), IT (50 hosts), HR (25 hosts), and a WAN link (2 hosts). Give the subnet + mask for each.

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Q42. From **10.0.0.0 /24**, you need three subnets: A (60 hosts), B (12 hosts), C (2 hosts). Provide subnet + mask for each.

Show / Hide Answer

Q43. In Q41, what goes wrong if you assign the **WAN /30 first** at 192.168.1.0, then try to fit Sales?

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Q44. From **172.16.0.0 /22**, allocate: Building1 (500 hosts), Building2 (250 hosts), Building3 (100 hosts). Provide subnet + mask for each.

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Q45. You subnetted **192.168.4.0 /24** with VLSM and used up to **192.168.4.223**. A new team needs **20 hosts**. What subnet + mask do you give them, and does it fit?

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SECTION 6 — Rapid-Fire Mixed Drills (Exam Simulation)

(No hints. Do these on a timer — target under 60 seconds each.)

Q46. **192.168.3.194 /26** → network, broadcast, first, last?

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Q47. **10.20.30.40 /28** → network, broadcast, first, last?

Show / Hide Answer

Q48. 172.31.100.100 /26 → network, broadcast?

Show / Hide Answer

Q49. 192.168.15.15 /29 → network, broadcast, first, last?

Show / Hide Answer

Q50. 10.0.5.130 /23 → network, broadcast? (Third-octet subnetting!)

Show / Hide Answer

Q51. 192.168.1.222 /27 → is it a host, network, or broadcast?

Show / Hide Answer

Q52. 172.16.14.0 /23 → how many usable hosts, and what's the broadcast?

Show / Hide Answer

Q53. 10.10.10.100 /25 → which subnet (network address)?

Show / Hide Answer

Q54. 192.168.100.100 /30 → network, broadcast, both usable hosts?

Show / Hide Answer

Q55. You see 255.255.255.252. How many of these subnets fit in a /24, and what are they good for?

Show / Hide Answer

SECTION 7 — Boss Level: Full Scenario Problems

(These mimic the harder simulation-style exam questions. Take your time.)

Q56. A router interface is configured with 192.168.1.65 /26. A host on that LAN is set to 192.168.1.130 /26 with gateway 192.168.1.65. The host can't reach the internet. What's the problem and the fix?

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Q57. You're given 192.168.50.0 /24 and must support **6 subnets**, each needing **25 usable hosts**. Give the mask and list all 6 network addresses.

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Q58. 10.1.1.0 /24 needs these, biggest-first (VLSM): Subnet A 120 hosts, B 60 hosts, C 30 hosts, D 12 hosts, plus two /30 WAN links (E, F). List each subnet + mask + range.

Show / Hide Answer

Q59. Summarize (aggregate) these four networks into ONE route: 172.16.0.0/24, 172.16.1.0/24, 172.16.2.0/24, 172.16.3.0/24.

Show / Hide Answer

Q60. Summarize 192.168.8.0/24 through 192.168.15.0/24 into one route.

The One-Page Speed Method (Print This!)

When you see any "IP + CIDR" question, do this in your head, every time:

STEP 1: Which octet is "interesting"?

/1–/8 → 1st octet

/9–/16 → 2nd octet

/17–/24 → 3rd octet

/25–/32 → 4th octet

STEP 2: Block size = $256 - (\text{mask value in that octet})$

STEP 3: Count 0, block, 2×block... until you pass the IP.
The one at/just-below = NETWORK.

STEP 4: Broadcast = next network – 1
First host = network + 1
Last host = broadcast – 1

STEP 5: Usable hosts = $2^{(32-\text{CIDR})} - 2$

Final Tips

1. **Memorize the cheat table** (/24–/30 with block sizes and host counts). It saves you 30 seconds every question.
2. **Always find the block for BOTH addresses** when asked "same subnet?" — don't eyeball it.
3. **Watch out for network & broadcast addresses** disguised as host answers (a favorite trick).
4. **VLSM golden rule:** biggest subnet first, every time.
5. **Do 10 problems a day.** Speed comes only from reps. In two weeks this will feel easy.

You've got this. Now go drill!

— End of Subnetting Drill Sheet —